Student Performance Prediction Using Machine Learning

Project report: binary classification of low final mathematics grades

1. Objective

This project compares Logistic Regression, Random Forest, and XGBoost for identifying students at risk of receiving a low final mathematics grade. The target is not student dropout. It is defined as low_grade = 1 when G3 < 10 and 0 otherwise.

2. Dataset

The project uses the mathematics subset of the UCI Student Performance dataset: 395 records and 33 columns. Target counts are 265 students with G3 >= 10 and 130 with G3 < 10. The input feature matrix excludes G3 and the derived target, leaving 32 features. The initial missing-value check found no pandas-recognized missing values.

3. Methodology

An 80:20 train/test split was used with random_state=42 and stratification: 316 training records and 79 test records. Numeric features were processed using StandardScaler and categorical features using OneHotEncoder(handle_unknown='ignore'), combined with models in a scikit-learn Pipeline. Models were Logistic Regression (max_iter=1000), Random Forest (100 trees), and XGBoost (100 estimators, max_depth=3, learning_rate=0.1); a most-frequent DummyClassifier served as baseline. Metrics were accuracy, recall, F1-score, and ROC-AUC. Five-fold stratified cross-validation was also run on the training data.

4. Test-set results

Model / experiment	Accuracy	Recall	F1	ROC-AUC
Logistic Regression (G1/G2 included)	93.67%	84.62%	89.80%	97.68%
Random Forest (G1/G2 included)	91.14%	88.46%	86.79%	97.50%
XGBoost (G1/G2 included)	91.14%	84.62%	86.27%	97.46%
Logistic Regression (G1/G2 excluded)	69.62%	30.77%	40.00%	71.48%
Most-frequent baseline	67.09%	0.00%	0.00%	Not reported

The Logistic Regression confusion matrix was [[52, 1], [4, 22]] for class 1 = low grade: 52 true negatives, 1 false positive, 4 false negatives, and 22 true positives.

5. Five-fold cross-validation (mean ± standard deviation)

Model	Accuracy	Recall	F1	ROC-AUC
Logistic Regression	0.905 ± 0.042	0.866 ± 0.114	0.854 ± 0.072	0.970 ± 0.017
Random Forest	0.902 ± 0.044	0.876 ± 0.103	0.853 ± 0.070	0.960 ± 0.030
XGBoost	0.915 ± 0.028	0.876 ± 0.071	0.871 ± 0.042	0.967 ± 0.021

XGBoost had the highest mean cross-validation accuracy and F1-score; Logistic Regression had the highest mean ROC-AUC. Random Forest and XGBoost had the same mean recall. These results do not establish a universally best model.

6. Ablation: excluding G1 and G2

For Logistic Regression, removing G1 and G2 reduced test accuracy from 93.67% to 69.62%, recall from 84.62% to 30.77%, F1 from 89.80% to 40.00%, and ROC-AUC from 97.68% to 71.48%. Cross-validation without G1/G2 was accuracy 0.674 ± 0.059, recall 0.338 ± 0.104, F1 0.401 ± 0.088, and ROC-AUC 0.633 ± 0.047. This suggests earlier grades carry substantial predictive signal; it does not demonstrate causation.

7. Conclusion

All three models performed strongly on the held-out test set when earlier grades were included. XGBoost led on mean cross-validation accuracy and F1-score, while Logistic Regression led on mean ROC-AUC. Performance dropped substantially without G1 and G2. The project is exploratory and predicts a derived low-grade label, not actual dropout.

8. Limitations

The dataset is small and represents one educational context. The target is derived from G3, not actual dropout. The availability of G1 and G2 depends on the intended prediction time. The results are predictive associations, not causal evidence, and there is no external validation. Future work should test larger, more diverse datasets, specify prediction timing, examine subgroup performance, and validate externally.

Reproducibility note: figures above are based on results recorded during the notebook audit. Re-run the notebook if code or data are changed.